

Ryan Vuong

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EXPERIENCE

Software Development Engineer

July 2023 – Present

Amazon Web Services

Santa Clara, CA

- Owned and operated components of a large-scale distributed system vending authoritative networking data for all AWS data centers worldwide.
- Designed and implemented an SNS/SQS-based eventing pipeline for near real-time data propagation, reducing API call volume by 15%
- Built infrastructure to transform unstructured network entity data into optimized data models, enabling efficient API access and storage for 1B+ daily requests from internal AWS services.
- Extended enforced data schemas with historical audit metadata, enabling efficient storage and querying of entity change history for improved data integrity and analysis.
- Built an internal chatbot using AWS Strands and Bedrock, enabling secure, real-time querying of proprietary networking data via an internal MCP server.
- Designed and implemented a CLI to query networking data across 50+ AWS regions concurrently, reducing investigation and operational workflows from hours to minutes.
- Owned on-call responsibilities and conducted service audits to support global operations, resolving high-impact incidents and reducing recurring issues by 65%.
- Led a team-wide migration from long-term IAM user credentials to short-lived IAM role credentials (6-hour rotation), eliminating persistent secrets and reducing blast radius from credential exposure.
- Mentored engineers via recorded learning sessions and regular 1:1s, improving onboarding efficiency and raising team-wide technical proficiency.

Software Development Engineer Intern

June 2022 – September 2022

Amazon Web Services

Seattle, WA

- Optimized full IPv4 scan of the internet, increasing the speed by 8x.
- Processed terabytes of IPv4 scan data using PySpark on AWS EMR to identify unresponsive address ranges.
- Automated large-scale data processing and storage using AWS Glue, S3, and Java, increasing IP probing response rates by 15%.
- Optimized AWS EC2 infrastructure by removing extraneous instances, resulting in 60% cost savings.

EDUCATION

University of California - Los Angeles

Los Angeles, CA

B.S. Computer Science

Sep. 2019 – June 2023

GPA: 3.76

Relevant Coursework: Operating Systems, Data Structures, Networks, Machine Learning,

TECHNICAL SKILLS

Languages: Python, Java, Ruby, SQL, JavaScript/TypeScript

Cloud & Infrastructure: AWS (DynamoDB, Lambda, S3, SNS, SQS, CloudWatch), CDK, Docker, IAM

Systems Experience: Distributed Systems, Data Modeling & Schema Design, Scalability & Performance Optimization

PROJECTS

nEARby | *Dart, Flutter*

- Developed a full-stack mobile application using Flutter which connects nearby users together using Spotify history
- Connected user profiles from Spotify utilizing the Spotify API; created internal user profiles within the app.

Toasty | *C#, Git, Unity*

- Lead a team of 7 developers to create and publish a game.
- Utilized object-oriented principles, inheritance, and polymorphism to develop player and enemies. Implemented 2D movement.